

Sensitivity of voice-onset time (VOT) measures to certain segmental features in speech production

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Abstract:

Voice-onset time (VOT) associated with the initial consonant of CVC words was measured as certain features of the vowel and final consonant were varied systematically. Specifically the experimental manipulations included varying of the tense/lax distinction of vowels and the voicing characteristic of the final consonant. Three subjects recorded each of 36 test words five times, as spoken in the carrier frame *Say ——— instead*. VOT and vowel duration measurements were obtained from spectrographic displays, and showed that (1) VOT was longer when the vowel was tense, as compared to lax, regardless of the voicing characteristic of the final consonant, (2) VOT was longer when the final consonant was voiced, as compared to voiceless, and (3) the VOT difference between words in which the vowel is tense and final consonant voiced, as compared to those in which the vowel is lax and final consonant voiceless, is an additive function of the effects noted in (1) and (2) above. Further analysis failed to reveal a meaningful covariance between VOT and vowel duration, so the feasibility of linguistic and/or physiological explanations for these data is discussed.

Introduction

In their important works dealing with context effects on voice-onset time (VOT), Lisker & Abramson (1967) and Klatt (1975) identified several factors which may exert systematic influence on this measure of glottal-supraglottal timing. For example, these studies have determined that VOT is conditioned by such factors as stress, nature of speech samples (i.e. sentential *vs.* citation material), and certain phonetic contexts which may include consonant cluster effects and the identity of the segment preceding the stop for which VOT is to be measured.

One inconsistency in the Lisker & Abramson (1967) and Klatt (1975) studies concerns the effect of vowel identity on the VOT associated with a preceding stop. Based on an experiment in which VOT was measured for /b, d, g, p, t, k/ in combination with twelve syllabic nuclei, Lisker & Abramson (1967, p. 15) concluded that vowel identity had no systematic influence on VOT (but see their footnote 16). But, Klatt (1975, p. 691) reported a statistically significant difference in VOT values for voiceless stops, depending on whether a high- or mid-vowel followed the stop. Klatt suggested that the failure of the Lisker and Abramson study to detect this vowel effect may be related to the limited scope of the former authors' data.

It is possible, however, that certain aspects of the two sets of data described above may reflect influences not previously considered in the literature. For example, Klee, Weismer & Ingrisano (1976) reported on a limited set of data in which VOT associated with the first consonant in a CVC word was altered systematically by the voicing characteristic of the *final* consonant. In addition, the Klee *et al.* data appeared to be in conflict with the trend of vowel conditioning on VOT found by Klatt, in that a low vowel (/æ/) yielded longer VOT's than a high vowel (/ɪ/). Upon further examination, it appeared that part of this discrepancy might be explained by the types of vowel used in the two studies. Klatt's tongue-height contrast involved two high, tense vowels (/i/, /u/) and tense (/ay/) and lax (/ɛ/) mid-vowels, whereas the Klee *et al.* contrast was based on high (/ɪ/) and low (/æ/) lax vowels. Thus, it could be reasoned that an unidentified influence on VOT of the tense/lax distinction in vowels might produce the inconsistency in the findings of the two studies.

The possibility that the tense/lax distinction in vowels caused some systematic variance in VOT did not seem unreasonable. Tense/lax vowel pairs are associated with a duration contrast, and Klee *et al.* (1976) had previously identified an effect on VOT due to the voicing characteristic of the final consonants, which is also associated with a vowel duration contrast. The experiment reported below is an attempt to determine if the tense/lax vowel distinction exerts a systematic influence on VOT, and to replicate and define more accurately the effect on VOT of final consonant voicing characteristics reported by Klee *et al.*

Method

Subjects

Three of the nine subjects who participated in the Klee *et al.* study were employed in the present investigation. Two of these subjects spoke with mid-western-type dialects, whereas the remaining subject spoke with a dialect heard in southeastern Pennsylvania.

Speech samples

The speech samples were constructed to yield a set of CVC words in which the tense/lax and voiced/voiceless characteristics of the vowel and final consonant, respectively, were completely crossed. The initial consonant in each word was one of the three voiceless stop consonants /p/, /t/, /k/ of American English. The vowels considered as tense or lax follow the classification given by Ladefoged (1975, pp. 73–76), and include /i/, /ei/, /u/ and /ɪ/, /ɛ/, /æ/ as tense and lax vowels, respectively.

We did not control place of articulation within our manipulation of the voicing characteristic of the final consonant. Fischer-Jorgensen (1964) provides an excellent summary of research on the effect of place of articulation of a final consonant on preceding vowel duration, and concludes that no consistent trend emerges from the literature. The lack of necessity to control this factor allowed us to avoid the inclusion of CVC “nonsense sequences” in the word set. In certain cases where a CVC word could not be created with a particular vowel-voiced stop combination, a nasal was substituted for the stop, as in the word *tame*. This appeared to be a reasonable alternative, since Peterson & Lehiste (1960) have shown that nasals exert approximately the same effect on preceding vowel duration as do voiced stops. In addition, one word in the set is of the form CVCC (*kept*), since a /keC_{voiceless stop}/ word does not exist in English.

A set of 36 words was constructed using the guidelines just described. This word set can be thought of as incorporating nine subsets in which each subset contains four words

which completely cross the tense/lax and voicing characteristics of the vowel and final consonant, respectively. A representative subset which illustrates the comparisons of interest within the experiment is /pid/-/pɪg/-/pɪk/-/pɪk/. Three of the nine subsets confounded tongue height within the tense/lax distinction; a representative example of such a subset is /tub/-/tut/-/tæɡ/-/tæk/.

Procedures

A 36 card deck was prepared by printing each experimental word on a five-by-seven card. This deck was shuffled before presentation to each subject, who was instructed to speak each word in the carrier frame *Say — instead* five times at conversational rate. All speech samples were tape recorded using high quality equipment, and the 36 word protocol was completed by each subject in two sessions.

Data analysis

The measurements of interest for this study were derived from spectrographic displays of a total of 540 utterances (36 words $\times$ 5 repetitions $\times$ 3 subjects). These measurements included VOT associated with the initial voiceless stops of the test words and the duration of the vowel within the test word. Measurement of VOT was performed under the criterion stated by Klatt, who takes "... the sudden onset of vertical striations in the second and higher formants" (1975, p. 687) to represent the onset of voicing following the stop burst. Vowel duration was measured as the interval between the onset of glottal pulsing, as indicated by the criterion described above, and the final well-defined vertical striation preceding the closure interval associated with the final stop in the test word.

All measurements were made by the author in quasi-blind fashion, such that no formal identification of a particular test word was available at the time of measurement. In addition, an estimate of measurement errors was obtained by remeasuring sixty randomly selected test words from newly prepared spectrograms. This analysis indicated mean measurement errors of 2.4 and 2.1 milliseconds (ms) for VOT and vowel duration, respectively.

Results and Discussion

VOT data obtained in the present investigation are summarized in Tables I–III. All values entered in the tables are expressed in milliseconds, with the exception of those in columns labeled *N*, which indicate the number of cases on which the means and standard deviations are based.

Place of articulation

Group mean VOT values and standard deviations associated with the four possible combinations of vowel feature (i.e. tense/lax) and final consonant voicing (i.e. voiced/voiceless) are reported for each place of articulation in Table I. In addition, the set of data labeled *P* in the table shows the result of pooling across vowels and voicing conditions. When the four possible combinations of the vowel and final consonant voicing features are examined, it is seen that, with only one exception, VOT values decrease systematically from the tense/+ voice condition to the lax/– voice condition. That is, using the letter code given in Table I for the four conditions, the ranking of VOT magnitudes for each place of articulation is $A > B > C > D$. The exception is that the VOT for /t/ in the lax/+ voice condition (C) is greater than the VOT for /t/ in the tense/– voice condition (B). The mean

Table I Group mean VOT data and standard deviations as a function of place of articulation, for all possible combinations of the vowel tense/lax and final consonant voicing features, and for data pooled across all vowels and final consonants. Table code is as follows: P = data pooled across vowels and final consonants; T/+V = data for tense vowel and voiced final consonant (A); T/-V = data for tense vowel and voiceless final consonant (B); L/+V = data for lax vowel and voiced final consonant (C); L/-V = data for lax vowel and voiceless final consonant (D). The column labeled N indicates the number of cases on which the means and standard deviations are based. All values are expressed in milliseconds

Code		VOT	Std. dev.	N
P	p	52.83	14.59	180
	t	65.55	12.49	180
	k	69.71	13.11	180
T/+V	p	58.80	13.38	45
	t	69.51	11.03	45
	k	74.62	12.79	45
T/-V	p	55.64	13.34	45
	t	64.44	10.13	45
	k	70.31	14.13	45
L/+V	p	51.42	14.29	45
	t	68.35	14.74	45
	k	67.55	12.40	45
L/-V	p	45.46	14.24	45
	t	59.90	11.65	45
	k	66.35	11.87	45

VOT difference for each place of articulation between the tense/+ voice (A) and lax/- voice (D) conditions, or those conditions which show the largest absolute differences, is associated with a probability of occurrence¹ due to chance of less than 0.01. For this particular contrast data for individual subjects are consistent with the group trends without exception. It should also be noted that certain other contrasts within Table I appear to be statistically reliable, but these are discussed below with reference to general trends.

Isolated and combined effects of vowel and final consonant voicing on VOT

Sections A-E in Table II report group and individual subject mean and standard deviation data, pooled across place of articulation, for five contrasts which allow for evaluation of the isolated and combined effects of vowel tenseness/laxness and final consonant voicing on VOT. Absolute mean differences for each of the five contrasts are shown for the group data only. Sections A and B present data which indicate that mean VOT is longer when the vowel in the test word is tense, as compared to lax, regardless of the voicing characteristic of the final consonant. The group mean differences of 6.95 and 7.33 ms in sections A and B are associated with probabilities of occurrence due to chance of less than 0.002 and 0.01, respectively.

The effect on VOT of the final consonant voicing feature is reported in sections C and D of Table II. These data show mean VOT to be longer when the final consonant of the

¹ All statistical results reported in this paper are derived from *t*-tests, unless otherwise noted. Probability statements are based on an initial two-tailed hypothesis.

Table II Group and individual subject mean VOT data and standard deviations for five selected contrasts. Codes for the various combinations of vowel and final consonant features are the same as in Table I. All values are expressed in milliseconds

A. Comparison of tense/lax vowels with final consonant voiceless (Ex: /pik/ vs. /pɪk/)					B. Comparison of tense/lax vowels with final consonant voiced (Ex: /pid/ vs. /pɪg/)				
		VOT	Std. dev.	N		VOT	Std. dev.	N	
Group	T/-V	63.86	13.79	90	Group	T/+V	68.86	15.73	90
	L/-V	56.91	15.03	90		L/+V	61.53	16.20	90
	$\bar{X}$ diff =	6.95				$\bar{X}$ diff =	7.33		
Subj A	T/-V	67.86	14.19	30	Subj A	T/+V	77.86	13.26	30
	L/-V	67.86	9.90	30		L/+V	73.20	11.28	30
Subj B	T/-V	70.13	11.48	30	Subj B	T/+V	72.60	10.66	30
	L/-V	55.73	15.08	30		L/+V	62.00	16.60	30
Subj C	T/-V	53.60	9.19	30	Subj C	T/+V	56.13	14.28	30
	L/-V	47.13	11.99	30		L/+V	49.40	10.46	30
C. Comparison of final consonant voicing with tense vowel (Ex: /tid/ vs. /tik/)					D. Comparison of final consonant voicing with lax vowel (Ex: /kɪd/ vs. /kɪt/)				
		VOT	Std. dev.	N		VOT	Std. dev.	N	
Group	T/+V	68.48	14.37	120	Group	L/+V	61.75	15.81	120
	T/-V	64.61	13.79	120		L/-V	56.61	15.67	120
	$\bar{X}$ diff =	3.87				$\bar{X}$ diff =	5.14		
Subj A	T/+V	77.30	12.02	40	Subj A	L/+V	74.95	11.43	40
	T/-V	68.00	12.82	40		L/-V	68.50	10.21	40
Subj B	T/+V	70.45	10.68	40	Subj B	L/+V	58.80	15.16	40
	T/-V	72.40	11.76	40		L/-V	55.40	14.52	40
Subj C	T/+V	57.70	13.00	40	Subj C	L/+V	51.50	10.32	40
	T/-V	53.45	8.72	40		L/-V	45.95	13.08	40
E. Comparison of tense/lax feature of vowel and voicing of final consonant (Ex: /pid/ vs. /pɪk/)									
		VOT	Std. dev.	N					
Group	T/+V	69.00	16.04	90					
	L/-V	56.91	15.03	90					
	$\bar{X}$ diff =	12.09							
Subj A	T/+V	77.86	13.26	30					
	L/-V	67.86	9.90	30					
Subj B	T/+V	73.00	11.88	30					
	L/-V	55.00	15.08	30					
Subj C	T/+V	56.13	14.28	30					
	L/-V	47.13	11.99	30					

Mean values for certain combinations may be slightly different for different contrasts, due to certain words applying to certain contrasts, but not to others.

test word is voiced, as compared to voiceless. The mean differences of 3.87 and 5.14 ms are associated with probabilities of occurrence due to chance of less than 0.05 and 0.02, respectively.

Section E of Table II summarizes the combined effect on VOT of the tense/lax distinction in vowels and final consonant voicing features. Test words in which vowels are tense and the final consonant is voiced have a mean VOT for the initial consonant that is 12.09 ms longer than words in which the vowel is lax and the final consonant is voiceless. This mean difference is associated with a probability of occurrence due to chance of less than 0.002.

Finally, while the individual subject data contain a fair amount of variability with respect to the magnitude of VOT differences for the five contrasts in Table II, the direction of these differences is, in general, consistent with group data. The individual subject data show only one reversal of a group trend, that occurring for subject B in the T/+ V *vs.* T/-V contrast. And, subject A had identical values for both conditions of the T/-V *vs.* L/-V contrast.

Vowel tongue height

In order for this study to resolve the apparent inconsistency between the Klatt (1975) and Klee *et al.* (1976) findings on VOT and vowel tongue height, it would have to demonstrate an interaction on VOT of tongue height and the tense/lax feature of vowels. Table III contains mean VOT data presented according to vowel identity and place of articulation. The interaction discussed above does not appear to be present in these results; moreover, some general trends in these data do not appear to be consistent with those reported by Klatt or Klee *et al.* For example, if VOT values are examined for two similar tongue-height contrasts in which one contrast involves tense vowels (/i/-/ei/) and the other lax vowels (/ɪ/-/ɛ/), tongue height does not appear to exert a systematic influence on VOT, with the exception of values associated with /k/. Thus for Klatt's tongue-height data to be consistent with the present results, his tongue-height contrast would have to be biased with /kV —/ tokens. Table III also shows that the /u/ data are similar to those for /ei/.

The /ɪ/-/æ/ contrast in Table III is also inconsistent in general terms with the Klee *et al.* study, which demonstrated longer VOT's for stops paired with the low vowel. The present results show this tendency for /pV—/ sequences only, which precludes any general statement regarding vowel tongue-height influences on VOT.

We do not have a satisfactory explanation for the longer VOT for /k/ preceding high, as compared to mid-vowels, or for the longer VOT for /p/ preceding the low vowel /æ/, as compared to /ɪ/. It may be that the /k/ data reflects the fact that the transglottal pressure drop necessary for the initiation of voicing is realized more slowly when /k/ is paired with a high vowel, due to the unique pre-constriction volume conditions associated with dorsal stops.

General comments

This study has shown that when other factors are held constant, VOT of a word-initial stop is related systematically to features of the following vowel and final consonant in the word. Specifically, VOT will be longer when the vowel is tense, as compared to lax, and when the final consonant is voiced, as compared to voiceless. The absolute effect on VOT of the tense/lax vowel opposition is slightly greater than that due to the voiced/voiceless contrast for the final consonant. In addition, the combined effect of the former and latter, which is reported in section E of Table II, appears to be described best as a simple, or additive

Table III Group mean VOT data as a function of place of articulation, presented according to test-word vowel identity. All values are in milliseconds

	/i/		/ei/	
	VOT	N	VOT	N
p	57.33	30	56.73	30
t	67.06	30	67.86	30
k	79.40	30	67.40	30

	/ɪ/		/ɛ/	
	VOT	N	VOT	N
p	44.06	30	48.46	30
t	63.33	30	65.13	30
k	71.66	30	62.66	30

	/u/		/æ/	
	VOT	N	VOT	N
p	57.6	30	52.8	30
t	66.0	30	63.93	30
k	70.6	30	66.53	30

effect. This can be demonstrated by adding the group mean differences in sections A and D, or sections B and C, and comparing the sums to the combined effect given in section E.

In an attempt to explain these data, we first considered the most obvious possibility, that VOT was related in some systematic way to vowel duration. This seemed to be a reasonable *a priori* assumption, since the manipulation of the vowel and final consonant features both result in demonstrable influences on the duration of the vowel. Moreover, the demonstration of such a relationship would allow the independent effects on VOT of the tense/lax and voiced/voiceless manipulations to be explained by a common phenomenon. The results of our analysis, however, failed to support the hypothesis of a meaningful covariance between VOT and vowel duration. Specifically, we correlated VOT and the duration of the following vowel under two conditions. First, to determine if a general relationship existed between these variables, correlation coefficients were obtained within each of the four possible combinations of the vowel and final consonant voicing features. Secondly, correlation coefficients for VOT and vowel duration *difference scores* were obtained for the five contrasts reported in Table II. Within the resulting pool of correlations, the relationship of greatest absolute magnitude showed that no more than 20% of the VOT variance could be accounted for by fluctuations in vowel duration. We therefore conclude, in agreement with Klatt (1975), that vowel-context conditioned variability in VOT is not a function of vowel duration. It should be emphasized that the failure to establish such a relationship applies to the difference scores for VOT and vowel duration,

for any of the contrasts considered in Table II; that is, the systematic change in VOT for each of these contrasts is *not* predictable in a quantitative sense from the corresponding systematic change in vowel duration.

We can identify at least two alternate explanations for the present findings, one of which concerns only the tense/lax vowel effect, whereas the other could provide a simultaneous account of the vowel and final consonant effects. Hoard (1971) has proposed a phonological rule whereby an intervocalic voiceless stop is more tense when followed by a tense, as compared to lax, stressed vowel. Hoard equates the notion of increased plosive tenseness with greater degrees of aspiration, so the tense/lax effect may simply reflect the phonetic realization of this suggested phonological rule. It would be interesting to observe the effect on VOT of the tense/lax distinction in vowels that do not take primary stress, for Hoard's rule appears to predict that the effect obtained in the present experiment should be eliminated under these conditions.

A somewhat different explanation of the effect on VOT of final consonant voicing and the tense/lax vowel distinction would follow from the hypothesis that stressing strategies adopted by speakers are to some degree guided by the length of the vowel which is to be stressed. For example, stressing of shorter vowels may be accomplished with lesser articulatory force to avoid overshoot of stress-related physiological events. Such a reduction in articulatory force—which may reduce the “stressedness” of the vowel—could be anticipated at the stop preceding the stressed vowel, which in turn may result in shorter VOT. Very little is known about segmental effects on stressing strategies, but there is some indication from the literature that this is an area deserving of systematic study. Öhman (1967) has stated that the pitch rise accompanying a stressed syllable varies directly with syllable length; it is unclear, however, if Öhman is referring to an existing set of data, or if he is stating a hypothesis. In any case, the hypothetical conditioning of articulatory force by segment-specific stressing strategies would not appear to be a linguistically motivated characteristic of speech production, at least not in the sense of rule-governed phenomena (see above).

Finally, we are hesitant to ascribe perceptual import to the systematic, but small, differences obtained in the present investigation. For this reason, some type of physiological, rather than linguistic, interpretation would seem to be most appropriate for these data.

References

- Hoard, J. E. (1971). Aspiration, tenseness and syllabication in English. *Language* 47, 133–140.
- Klatt, D. H. (1975). Voice-onset time, frication, and aspiration in word-initial consonant clusters. *Journal of Speech and Hearing Research* 18, 686–706.
- Klee, T., Weismer, G. & Ingrisano, D. R. (1976). Laryngeal timing constraints in plosive-vowel and fricative-vowel sequences, or: Is VOT really the best measure of glottal-supraglottal timing? Paper presented to the 92nd meeting of the Acoustical Society of America, San Diego, CA.
- Ladefoged, P. (1975). *A Course in Phonetics*. New York: Harcourt Brace Jovanovich, Inc.
- Lisker, L. & Abramson, A. S. (1967). Some effects of context on voice onset time in English stops. *Language and Speech*, 19, 1–28.
- Öhman, S. E. G. (1967). Word and sentence intonation: A quantitative model. Quarterly Progress and Status Report, 2–3, 20–54: Speech Transmission Laboratory, Stockholm.
- Peterson, G. & Lehiste, I. (1960). Duration of syllabic nuclei in English. *Journal of the Acoustical Society of America*, 32, 698–703.